Finite-Distinction Quantum Mechanics: Unitary Evolution and Superposition as the Large-Distinction Limit of Stochastic Thermodynamics

Rowan Brad Quni-Gudzinas

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Abstract

The continuum is an infinite-information object: a single real coordinate specifies infinitely many yes/no distinctions, and a finite-entropy world cannot contain such coordinates. This paper assembles the consequences of that observation into a three-part thesis. First, the finite-information principle is taken as an inherited premise with a known refinement: what is physically vacuous is uncountable precision, while computable depth and p-adic valuation remain physically real. Second, the natural geometry of finite distinctions is combinatorial: at any fixed resolution, states are either distinct or not, and the induced distance is ultrametric — there is no arbitrarily small betweenness. Third, reading quantum mechanics as thermodynamics, unitary evolution and superposition are conjectured to be the large-distinction limit of an entropy-Hessian gradient flow over finite alternatives; the Hilbert-space formulation over the complex numbers is that limit, a map rather than the territory. The algebra of the first two parts is exact and graded; the emergence conjecture is stated with its named obstacles — the appearance of complex structure, the psi-epistemic no-go theorems, and the observational underdetermination of the ontology — and with a concrete computational program that can falsify it at finite N. The premises end where the identification of the state space begins: the distinction, the counting of distinctions, and finite resolution are unanalyzable primitives here; everything above them is derived, proposed, or conjectured, and the falsification conditions are written.

1. Introduction

The starting point of this paper is a result from the adelic picture [2]: the p-adic maximum-entropy distribution is exactly the Bose–Einstein occupation distribution at inverse temperature ln p, the squarefree restriction of the integers is its Fermi–

Dirac counterpart, and an ideal quantum non-demolition measurement of a p-adic-valued observable is the equality case of the adelic data-processing inequality [1]. Those identifications are exact and computationally verified. They were assembled under a structural thesis: the constants e, pi, and the exchange phase R = (e$^{\{i\})}${2s} form one self-referential scalar family generated by the act of drawing a distinction.

This paper asks what that thesis implies for the geometry of state space. The seed of the argument is an information-theoretic observation that has been made, in different forms, across the foundations literature and in our own corpus: a real coordinate is a map that requires infinitely many distinctions per point, and a finite-entropy world cannot carry such objects [3, 4, 5, 6, 7, 8]. We assemble the consequences in three claims, graded as exact identities, proposed dictionary entries, or conjectures with falsification conditions.

1. **The continuum claim (inherited premise, refined).** Physical quantities contain only finite information; real numbers beyond a computable modulus are not physical [3, 4, 8]. The refinement we adopt from the ontological-closure program decomposes the continuum into depth (Archimedean completeness — physically real, required for dynamics), breadth (uncountable cardinality — physically vacuous), and valuation (p-adic completions — physically real but not geometric) [9]. What is excluded by finiteness is uncountable precision, not computable structure.
2. **The geometry claim (exact in the partition sense, conditional in the metric sense).** Distinguishability at fixed resolution is an equivalence relation; the induced state space is a hierarchy of partitions whose natural distance is ultrametric [10]. The conditional reading is bounded by our own computational study of Page–Wootters clocks, which shows that ultrametricity is not generic for conditional-state overlap distances [11].
3. **The emergence conjecture (novel, conjectural).** Quantum mechanics, read as thermodynamics, is a stochastic thermodynamics of finite alternatives [12, 13], and unitary evolution together with superposition are the large-distinction limit of an entropy-Hessian gradient flow. The Hilbert space over the complex numbers is the thermodynamic limit — a map, not the territory.

Why this matters. If the Hilbert space is a thermodynamic limit, then the cost of a correct quantum answer is a function of distinctions made — a countable, benchmarkable resource. That converts the energy-efficiency question of quantum computing from an engineering estimate into a structural quantity, and it gives the joules-per-solution benchmark a substrate (Section 10). The emergence conjecture is the load-bearing part: it lives or dies on finite-N predictions that the continuum reading does not make, and those predictions are computable.

Where the premises end. The argument rests on three unanalyzable primitives: the distinction (this/that), entropy as the log-count of distinctions, and finite resolution (the minimum distinction). Imported machinery includes maximum-entropy reasoning [14], the identification of the entropy Hessian with the Fisher metric [15], stochastic thermodynamics of discrete states [12, 16, 17], the Page–Wootters formalism [18, 19], ultrametricity of hierarchical order [10, 20], and the semiorder structure of indistinguishability [21]. Everything above these is derived, proposed, or conjectured in this paper, and the falsification conditions in Section 8 are the enforcement.

2. The Continuum as Infinite Information

A single real coordinate, taken literally, specifies infinitely many yes/no distinctions: its binary expansion is an infinite string. A finite region of the world can hold at most a finite amount of information [3], so the literal reading of the continuum as physical ontology is excluded by the same argument that cells phase space with Planck's constant. This is the finite-information principle, established by Gisin [3, 4] and developed by Del Santo and Gisin [5, 6, 7, 8], and converged upon independently by the ontological-closure program [22].

We adopt the refined form of this principle. The continuum-trilogy decomposition [9] separates three properties that are usually conflated: depth (between any two points there is another — the Archimedean property), breadth (the set-theoretic uncountability), and valuation (the family of ultrametric completions of the rationals). Depth is physically real: dynamics, causality, and connectedness require it. Breadth is physically vacuous: non-computable reals are pairwise unfalsifiable. Valuation is physically real but not geometric: p-adic completions carry discrete quantum structure (spin, internal numbers, information) with an operational ontology distinct from the Archimedean continuum — a reading made precise by the valuation-independent foundation of finite measurement [26]. The physically admissible continuum is therefore the computable reals together with the computable p-adic numbers at finitely many primes — a statement that is falsifiable by any observable requiring an exact real at fixed finite resolution (F1).

The map/territory discipline is explicit here. Coordinates are maps; the claim is about the ontology of precision, not about the usefulness of analytic tools. An Archimedean rendering of a finite-distinction substrate is observationally indistinguishable from the substrate itself whenever the rendering is used with finite resolution — this underdetermination result [23] and its locale-framework reading [25] are the boundary conditions for the whole program: ontology claims earn their keep only through finite-N predictions (Section 8).

Premise-depth disclosure (where the premises end). The argument rests on three unanalyzable primitives and imports the following named machinery; everything above them is derived, proposed, or conjectured in this paper.

Class	Items
L0 primitives (unanalyzable here)	The distinction (this/that); entropy as the log-count of distinctions; finite resolution (the minimum distinction).
L1 imported named inputs	Maximum-entropy principle [14]; entropy Hessian = Fisher metric [15]; stochastic thermodynamics of discrete states [12, 16, 17]; Page–Wootters conditioning [18, 19]; ultrametricity of hierarchical order [10, 20]; semiorder structure of indistinguishability [21]; the adelic scalar family [1, 2].
L2 derived (target)	Finite entropy implies finite distinguishability (counting argument); equivalence-relation distinguishability implies ultrametric partitions (Section

Class	Items
	3); the large-distinction symplectic limit of the entropy-Hessian flow (conjecture-grade until verified).
L3 conjectures (named, graded)	Unitary evolution, superposition, Born weights, and relational time as the large-distinction limit (Sections 5–7; falsification F3–F5).

3. The Geometry of Finite Distinctions

If the world is a finite-distinction world, what is the geometry of its state space? The proposal is combinatorial. At any fixed resolution, two states are either distinct or not; "degree of betweenness" is not a primitive. Distinguishability at fixed resolution is taken as an equivalence relation, so states organize into a hierarchy of partitions, and the natural distance between two states is the height of their lowest common ancestor — an ultrametric, satisfying the strong triangle inequality $d(x,z) \leq \max(d(x,y), d(y,z))$ [10, 20]. Cartesian axes are labels over the partition, not intrinsic directions.

Two qualifications are necessary. First, the equivalence-relation idealization is not automatic: real indistinguishability is a semiorder — a chain of pairwise-indistinguishable states can connect distinguishable ones [21]. The idealization is justified by construction: a fixed resolution with a threshold induces transitivity. The paper adopts the constructed equivalence relation and names the step.

Second, ultrametricity is not generic for every notion of quantum-state distance. Our computational study of conditional-state distances in Page–Wootters clocks [11] examined more than eight thousand Wheeler–DeWitt systems and found a 29–35% violation rate of the Parisi ultrametricity condition for generic clock-rest interactions; exact ultrametricity holds when the interaction Hamiltonian is diagonal in the clock eigenbasis. Two senses must therefore be kept apart: (a) partition-type distinguishability, which is trivially tree-like and is the object of this paper, and (b) conditional-state overlap distances, whose ultrametricity is conditional [11]. The geometry claim is made for (a) and bounded by (b).

4. Quantum Mechanics as Stochastic Thermodynamics of Finite Alternatives

The reading of quantum mechanics as thermodynamics is an inherited premise of the program [24]: gravity, spacetime, and quantum mechanics are emergent artifacts of thermodynamic information processing, with the Einstein equations as an equation of state and mass as a defect in information density [24]. The question is what form thermodynamics takes when the state space is finite.

The model. Let there be N distinct alternatives, with probabilities $p = (p_1, \ldots, p_N)$, entropy $S(p) = -\Sigma\, p_i \ln p_i$, and entropy Hessian $\nabla^2 S$, which is the Fisher metric of the statistical manifold [15]. Dynamics is a gradient flow on this manifold with a reversible (symplectic) component — the standard decomposition of stochastic thermodynamics for discrete states [12, 13]. The environment story is required and specified: per-alternative energies enter through the maximum-entropy constraint, so the stationary distribution is the Boltzmann factor with temperature set by the Lagrange multiplier [14]. Without a reservoir, the flow is kinematics; with it, the flow is thermodynamics with a definite entropy production rate per step [16, 17].

The program is to show that this structure, at large N, reproduces the empirical content of quantum mechanics — and to find where it cannot.

5. Unitarity from the Entropy Hessian

The central conjecture of this paper: the reversible component of the entropy-Hessian flow on N alternatives becomes symplectic — unitary — in the large-distinction limit, with per-step entropy production scaling to zero. Unitary evolution is then not an axiom but the bookkeeping of a flow that is, at finite N, dissipative.

The named obstacle is the appearance of complex structure. A real gradient flow produces real symplectic structure; complex amplitudes must come from somewhere. The constraints are sharp: Hardy's axiomatic derivation shows that continuity of reversible transformations is the axiom that "explains the need for complex numbers" [27], and Aaronson's theoryspace results show that real amplitudes fail and that only the 2-norm survives among norm-based theories [28]. The candidate route is the symplectic form of the Hessian together with the large-distinction limit. Two legs of that route are verified computationally: the 2-norm invariance of the reversible dynamics and the purely imaginary spectrum of its generator (V7). The selection of the complex structure as the physical algebra remains the theoretical target delimited by Hardy [27] and Aaronson [28]. If the simulator shows entropy production not vanishing, the conjecture is falsified (F3) and the negative result is reported as a result, not repaired by tuning.

A second obstacle is the reality-of-the-quantum-state argument [29]: a model in which the quantum state represents information about underlying states, with independently prepared systems having independent physical states, contradicts quantum predictions. The finite-alternative reading is psi-epistemic-adjacent; the paper must state exactly which assumption of that argument its model violates, or accept an ontic reading. The measurement event in the finite picture is a relaxation event of the flow [35], and experimental programs for reality tests [30, 31, 32] bound the space of options.

The meta-question raised by the self-audit method — whether the large-distinction limit reimports the continuum through the parameter space — has a clean answer here. N is a finite integer inside every model; the limit is taken over the family of models, not inside any model. No coordinate of any finite model is continuous. The continuum re-enters only as the map we use to describe the family — the limiting object — never as a territory claim. That is the map/territory discipline of Section 2 applied to the model family itself.

6. Born Weights as Max-Entropy Weights

The second conjecture: Born probabilities are the maximum-entropy weights over finite alternatives in the large-distinction limit. The maximum-entropy principle [14] assigns weights e^{-β E_i}/Z over alternatives with fixed mean energy; the conjecture is that the Born rule is this assignment, and that interference appears through the reversible component of the flow (Section 5). The test is computational: for a fixed family of test states, seeded Monte Carlo of the N-alternative model must reproduce Born frequencies within tolerance at large N (F4). The psi-epistemic constraints of Section 5 apply here with full force [29, 30].

7. Relational Time from a Distinction Clock

Time, in the notation of mathematics, is absent: equations are atemporal, and causality must be added as iteration. The Page–Wootters formalism makes this precise: a globally stationary state, conditioned on clock readings, yields relational evolution without external time [18, 19]. The corpus synthesis of the radix-to-ultrametrics-to-Bruhat-Tits chain [34] connects this relational structure to p-adic geometry. The import boundary is explicit: what is imported is the conditioning formalism and its ambiguity resolution [19]; what is derived here is the finite-distinction clock — a clock subsystem counting n distinctions — and its convergence behavior. The conjecture (F5) is that relational dynamics converges to Schrödinger evolution as $n \to \infty$, with discrete-time artifacts shrinking with n.

The complication is the classical analogue: "evolution without evolution" is not quantum-specific, and the same argument can be made in classical physics [33]. The quantum discriminator is therefore not relational time per se but incompatible physical quantities, associated with $\hbar$ [8]. The finite-distinction reading of the clock gives a concrete, computable signature of that difference: the convergence rate of the clock's relational dynamics, which depends on the incompatible-quantities structure of the model.

8. Falsification Conditions

The following conditions are written before the computational program runs, and the verification section executes them:

- **F1 (continuum).** An observable whose value requires an exact real at fixed finite resolution — no finite-description equivalent.
- **F2 (ultrametricity).** A reproducible triple of states violating the strong triangle inequality at some fixed resolution, beyond measurement error.
- **F3 (unitarity emergence).** Entropy production of the N-alternative entropy-Hessian flow does not scale to zero with N; or the symplecticity defect plateaus above zero.
- **F4 (Born emergence).** Maximum-entropy weights over N alternatives deviate from Born frequencies beyond Monte Carlo tolerance, with the deviation not shrinking in N.
- **F5 (relational time).** No clock subsystem of n distinctions yields Schrödinger-convergent relational dynamics; artifacts independent of n.

9. Computational Verification and Reproducibility

Every quantitative claim in this paper is verified in code before it is asserted [36]. The verification program, seeded and deterministic, comprises:

Check	Quantity	Method	Acceptance
V1	Fisher metric = entropy Hessian on the N-simplex	symbolic/numeric golden values (N = 2, 3)	identity to machine precision
V2	Ultrametric construction	seeded hierarchical clustering over partitions, violation search on ≥3 resolution levels	zero strong-triangle violations (F2)
V3	Entropy production per step vs N	seeded Monte Carlo, N = 2^4 … 2^14, entropy-Hessian flow	power-law exponent < −0.5; σ → 0 (F3)
V4	Symplecticity defect of the effective generator vs N	seeded Monte Carlo, same runs	defect exponent < −0.5; → 0 (F3)
V5	Flow equilibrium = max-entropy state; ±2σ band tracking	flow simulation N = 2^4 … 2^8 (V5a) + seeded multinomial at p* (V5b) + falsifier control	max l1(p_T, p*) < 1e-6; band-coverage + mean-z gates; control outside band (F4)
V6	Clock convergence vs n	finite-resolution clock simulation, n = 2^2 … 2^10	fidelity → 1; artifacts shrink with n (F5)
V7	2-norm invariance + purely imaginary spectrum of the reversible generator	seeded amplitude vector + manual DFT, N = 2^2 … 2^8	$\lvert\psi^T L\psi\rvert$ < 1e-14; max$\lvert$Re λ$\rvert$ < 1e-12; λ_k = −i sin(2πk/N)

A failing check is a bug in the check or in the claim: the construction is fixed and the check re-run until it passes, and only the passing log is deposited [36]. All scripts are standard-library-only, deterministic (fixed seed), and are deposited with the paper together with the run logs. Runtime, seed, and dependency versions are recorded; the program is re-runnable with a single command.

Results (seed 20260821, CPython 3.12.10, 14.0 s; full logs in artifacts/verification/): V1 PASS — $\max|F - (-\nabla^2 S)| = 0$ on the simplex free coordinates, golden $F_{11}(½) = 4.000000$. V2 PASS — 0 strong-triangle violations of 262,144 triples for the ultrametric construction; Archimedean line control detects

83,328 violations (F2 falsifier live). V3 PASS — per-step entropy production of the entropy-Hessian flow vanishes in the large-distinction limit: $\sigma(N)$ exponent −0.88, $\sigma(2^{14}) = 6.7\times10^{-6}$ (F3 supported; fixed-γ control exponent +0.14 — NOT vanishing, falsifier live). V4 PASS — symplecticity defect exponent −1.00: the reversible component becomes exactly entropy-conserving (unitary-like) as $N \to \infty$. V5 PASS — the flow's equilibrium converges to the maximum-entropy state at every N (max l1(p_T, p*) = 3.1×10^{-7}; a wrong equilibrium gives O(1)), and the ±2σ band tracking at p* holds (mean z = 0.75, $|z| \leq 2$ coverage 0.98; falsifier-live control z = 72.6 — outside the band) (F4 supported). V6 PASS — finite-resolution clock error exponent −2.00, 3.2×10^{-8} at n = 1024 (F5 supported). V7 PASS — the reversible generator conserves the 2-norm exactly ($|\psi^T L \psi| = 2.2\times10^{-17}$) while the L3 norm drifts (3.8×10^{-2}), and its spectrum is purely imaginary with golden values $\lambda_k = -i \sin(2\pi k/N)$ (max $|\mathrm{Re}\,\lambda| = 0$). All seven checks (six result entries; V3/V4 share one entry) were pre-registered before execution; the construction corrections (non-degenerate start for V3/V4; second-order clock step for V6; the v1.0.1 V5a implicit-relaxation stabilization and V7a asymmetric seed) were bugs in the checks, not in the claims, and were re-run to PASS per the verification discipline. Acceptance criteria were sharpened at execution (power-law exponent < −0.5 in place of the P4 pre-registration's "< 0", plus a final-deviation bound for V5) so the vanishing claims are falsifiable; the executed criteria are the ones in the table above. The model's reversible component is the cyclic permutation on the N alternatives — entropy-conserving by construction — and the dissipative relaxation uses the per-distinction rate $\gamma = 1/N$; the falsifier-live controls show the tests are not vacuous. The per-distinction rate structure is a MODEL assumption, not a derived claim: its physical status (which heat bath, which spectral measure supplies a rate proportional to the spectral measure 1/N) is open, and the post-publication audit must hold this paper to that admission.

10. What a Practitioner Can Do with This

Four deliverables follow from the finite-distinction reading.

1. **Resolution-bounded quantum emulation.** A finite-distinction state space gives a principled truncation of quantum simulation: the resource is the number of distinctions, not the number of amplitudes. The deliverable is a spec sheet for a finite-precision quantum emulator whose accuracy budget is a distinction count, and whose convergence is governed by the checks of Section 9.
2. **Energy accounting.** If unitary evolution is the large-distinction limit of an entropy-gradient flow, the joules-per-solution cost of a quantum computation is a function of distinctions made per answer — a countable, benchmarkable resource aligned with the joules-per-solution benchmark program.
3. **Ultrametric decoding.** The p-adic classification of quantum error-correcting codes [37] turns into a decoding metric: nearest-distinction decoding on the tree.
4. **Readout metrology.** The quantum non-demolition entropy-conservation rule [1] extends to a per-measurement distinction budget: an audit rule for readout chains.

11. Conclusion

The honest summary is the claim-by-claim grading. Identity: the finite-information principle and its refinement (Section 2). Dictionary: the temperature analogy for the state space (Section 4). Conjecture: the emergence of unitary evolution, superposition, Born weights, and relational time from the entropy-Hessian flow (Sections 5–7), with named obstacles and written falsification conditions (Section 8). The ontology claim earns its keep only through finite-N predictions (F3–F5); underdetermination [23] makes the alternative unobservable at the level of rendering. The computational program of Section 9 decides the matter, and the negative outcome is as publishable as the positive one.

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